

VPPL SOLAR & WIND ENERGY SCHEDULING SYSTEM

INTELLIS DAY-AHEAD LOGIC
ARCHITECTURE

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Document Revision History & Implementation Mandate

Revision	Release Date	Primary Author	Detailed Summary of Modifications & Architectural Scope
v1.0	2026-10-08	Chaitanya Gore	Initial Complete Production Specification for Day-Ahead Scheduling Architecture. Formulates end-to-end low-level algorithmic logic for both Solar and Wind Day-Ahead (T+1) scheduling. Key architectural pillars include: (1) Strategic elimination of generative LLM dependencies in favor of deterministic pure physics and multi-agency Model Output Statistics (MOS) consensus; (2) Statutory gate closure compliance for 10:00 AM IST under Indian Electricity Grid Code (IEGC 2023) across 96 daily dispatch blocks; (3) Astronomical NREL SPA and Ineichen clear-sky ephemeris with Hay-Davies POA transposition; (4) 143-member NWP super-ensemble ingestion across ECMWF IFS, GFS, ICON, and GEM with physics-guided Kt clearness splines; (5) Faïman and Sandia convective cell temperature simulation with dynamic thermal derating; (6) Diurnal 3-slot ensembling with mandatory agency diversity capping (maximum 2 models per NWP agency) and Bayesian inverse-loss weighting; (7) DynamoDB planned maintenance outages and control window capacity scaling; (8) Multi-generator asset decomposition with dedicated column preservation (e.g. ZTRIC, ENRICH, SHAHA); (9) Wind aerodynamic turbine profiling, IEC 61400-12 dynamic air density normalization, Hellmann diurnal atmospheric boundary layer shear, and Jensen's-inequality-safe member-by-member power transformation; (10) Four physical guardrails; and (11) Canonical statutory 96-block CSV formatting and multi-path Amazon S3 dispatch.

Codebase Directory & Implementation File Tree Structure

The algorithmic logic, mathematical models, and data ingestion pipelines specified within this document are implemented across the following structured module repository hierarchy under schedule/. Each component is isolated for serverless AWS Lambda execution with zero cross-site dependency.

schedule/	
├─ intellis_dayahead_lambda.py	[Centralized Solar DA Lambda Entrypoint]
├─ intellis_dayahead_wind_lambda.py	[Centralized Wind DA Lambda Entrypoint]
├─ deploy_dayahead_lambda.py	[Solar DA Deployment & Rollback Script]
├─ deploy_dayahead_wind_lambda.py	[Wind DA Deployment & Rollback Script]
├─ modules/	
│ └─ day_ahead_forecast_solar/	
│ └─ solar_da_engine.py	[Master Solar Day-Ahead Orchestrator]
│ └─ da_clearsky_engine.py	[NREL SPA Ephemeris & Ineichen Clear-Sky]
│ └─ da_nwp_spline_engine.py	[143 NWP Ensemble Ingestion & Kt Splines]
│ └─ da_physics_baseline.py	[Faïman Cell Temp & Thermal Derating]
│ └─ da_member_selection.py	[3-Slot Diurnal Ensembling & Agency Cap]
│ └─ da_maintenance_derate.py	[DynamoDB Day T+1 Outage & Derate Scaling]
│ └─ da_guardrails_engine.py	[Physical Clamping & 0.01 MW Rounding]
│ └─ da_formatter_s3.py	[Statutory 96-Block S3 CSV Formatter]
│ └─ day_ahead_forecast_wind/	
│ └─ wind_da_engine.py	[Master Wind Day-Ahead Orchestrator]

		— wind_density_iec_engine.py	[IEC 61400-12 Air Density Normalization]
		— wind_nwp_hub_engine.py	[Hellmann Boundary Shear & 15-Min Splines]
		— wind_aerodynamic_power.py	[Jensen's-Safe OEM Power Curve Lookup]
		— wind_park_derate_engine.py	[Combined 3-Factor Park Derate Engine]
		— wind_formatter_s3.py	[Statutory 96-Block S3 CSV Formatter]

Implementation File Mapping & Component Responsibilities:

Directory / Path	Code File Name	Architectural Role	Core Logic & Functions Implemented
schedule/	intellis_dayahead_lambda.py	Solar DA Entrypoint	lambda_handler(), EventBridge DA0/DA1 trigger parsing, multi-plant dispatch loop, S3 upload coordination.
schedule/	intellis_dayahead_wind_lambda.py	Wind DA Entrypoint	lambda_handler(), EventBridge wind trigger parsing, 24h continuous wind scheduling (no night zeroing).
schedule/modules/day_ahead_forecast_solar/	solar_da_engine.py	Master Solar DA	End-to-end 96-block pipeline orchestration across all solar sub-engines.
schedule/modules/day_ahead_forecast_solar/	da_clearsky_engine.py	Ephemeris Engine	NREL SPA algorithm, true solar time, zenith angle, Ineichen clear-sky irradiance profile.
schedule/modules/day_ahead_forecast_solar/	da_nwp_spline_engine.py	NWP Spline Engine	Open-Meteo 143-member ingestion, hourly clearness index (Kt), 96-block cubic Hermite splines.
schedule/modules/day_ahead_forecast_solar/	da_physics_baseline.py	Thermal Derating	Faiman cell temperature equation, temperature coefficient gamma, DC array conversion.
schedule/modules/day_ahead_forecast_solar/	da_member_selection.py	Diurnal Ensembling	3-Slot diurnal regime partitioning, agency diversity capping (≤ 2 members per agency).
schedule/modules/day_ahead_forecast_solar/	da_maintenance_derate.py	Outage Scaling	DynamoDB plant_control_window

			s1 query for Day T+1 maintenance windows, capacity derating.
schedule/modules/day_ahead_forecast_solar/	da_guardrails_engine.py	Grid Guardrails	Inverter clipping ($0 \leq P \leq P_{AC,max}$), night-time zeroing, 0.01 MW precision rounding.
schedule/modules/day_ahead_forecast_solar/	da_formatter_s3.py	S3 Lake Delivery	96-block statutory table assembly [S. No, From, To, SCH_MW, AvC_MW], S3 upload to intellis Dayhead solar/.
schedule/modules/day_ahead_forecast_wind/	wind_da_engine.py	Master Wind DA	End-to-end 96-block wind scheduling across all aerodynamic and thermodynamic modules.
schedule/modules/day_ahead_forecast_wind/	wind_density_iec_engine.py	Air Density IEC	IEC 61400-12 air density calculation, elevation and temperature normalization.
schedule/modules/day_ahead_forecast_wind/	wind_nwp_hub_engine.py	Boundary Layer	Hellmann power-law shear elevation from 10m to 100m-140m hub heights, 15-min splines.
schedule/modules/day_ahead_forecast_wind/	wind_aerodynamic_power.py	Jensen's Safe Curve	Non-linear member-by-member OEM power curve evaluation prior to ensemble averaging.
schedule/modules/day_ahead_forecast_wind/	wind_park_derate_engine.py	Park Derate Engine	3-Factor park derate ($\eta_{park} = \eta_{wake} * \eta_{elec} * \eta_{avail}$ approx 0.89).
schedule/modules/day_ahead_forecast_wind/	wind_formatter_s3.py	S3 Lake Delivery	96-block wind statutory CSV assembly, S3 upload to intellis Dayhead wind/.
schedule/	deploy_dayahead_lambda.py	Solar Deployment	Automated deployment, container image update, and rollback verification.

schedule/	deploy_dayahead_wind_lambda.py	Wind Deployment	Automated deployment, container image update, and rollback verification.
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1. System Architecture & Day-Ahead Regulatory Governance Overview

The Intellis Day-Ahead Scheduling System is an automated, high-precision forecasting and grid commitment platform engineered for utility-scale solar photovoltaic power stations and wind turbine generation fleets across India. The architecture complies strictly with the statutory mandates of the Indian Electricity Grid Code (IEGC 2023) and guidelines established by the Central Electricity Regulatory Commission (CERC) and State Electricity Regulatory Commissions (SERCs) including Telangana (TSERC), Madhya Pradesh (MPERC), and Maharashtra (MERC).

1.1 Regulatory Mandate & The 10:00 AM Gate Closure

Under national grid operating codes, renewable energy generators (wind and solar) connected to the Inter-State Transmission System (ISTS) or State Transmission Utility (STU) networks are legally required to submit Day-Ahead generation schedules for the subsequent operational date (Day T+1). The statutory deadline enforced by Regional Load Despatch Centres (RLDCs) and State Load Despatch Centres (SLDCs) is gate closure at 10:00 AM IST on Day T (with certain state jurisdictions permitting submissions up to 11:30 AM IST).

Day-Ahead schedules submitted by generators form the foundation for national power grid commitment, transmission corridor congestion management, inter-regional power flow scheduling, and day-ahead power market clearing on energy exchanges (IEX, PXIL, HPX). Because the initial Day-Ahead schedule establishes the statutory baseline against which Deviation Settlement Mechanism (DSM) commercial penalties are evaluated, schedule accuracy is critical to plant financial viability.

1.2 Day T+1 15-Minute Time Block Discretization

Day-Ahead schedules must specify generation commitments for every 15-minute time block across the entire 24-hour dispatch horizon of the target date (Day T+1), spanning exactly ninety-six (96) continuous blocks indexed b in $\{1, 2, \dots, 96\}$.

$$D_{\text{target}} = D_{\text{today}} + 1 \text{ day}, t_{\text{start}}(b) = (b - 1) \times 15 \text{ min}, t_{\text{end}}(b) = b \times 15 \text{ min}, b \in \{1, 2, \dots, 96\}$$

Where: Standard 96-block Day T+1 grid temporal discretization.

Block Index (b)	Time Interval (IST)	Operational Context (Solar)	Operational Context (Wind)	Statutory Significance
1	00:00 - 00:15	Deep Night (GTI = 0, P = 0.00 MW)	Nocturnal Boundary Layer Jet Active	Day T+1 First Dispatch Block

24	05:45 - 06:00	Solar Astronomical Horizon Threshold	Morning Inversion Dissipation Window	Solar Generation Window Opens
25	06:00 - 06:15	Morning Solar Ramp (Slot 1)	Morning Boundary Layer Breakdown	Active Daylight Injection Begins
41	10:00 - 10:15	Midday Solar Peak Window (Slot 2)	Convective Daytime Lull Window	Peak Solar Irradiance Begins
56	13:45 - 14:00	Midday Solar Peak Window (Slot 2)	Convective Daytime Lull Window	Maximum Thermal PV Derating
57	14:00 - 14:15	Afternoon Solar Decay (Slot 3)	Late Afternoon Thermal Breeze	Afternoon Solar Decline Begins
74	18:15 - 18:30	Solar Astronomical Horizon Threshold	Evening Radiative Inversion Ramp	Solar Generation Window Closes
75	18:30 - 18:45	Deep Night (GTI = 0, P = 0.00 MW)	Evening Low-Level Jet Formation	Nocturnal Hard Zeroing Enforced
96	23:45 - 24:00	Deep Night (GTI = 0, P = 0.00 MW)	Stable Nocturnal Boundary Jet	Final Dispatch Block of Day T+1

1.3 Strategic Elimination of Generative LLM in Day-Ahead Scheduling

A critical architectural evolution in the Intellis system is the complete elimination of generative Large Language Models (LLMs) in the Day-Ahead scheduling pipeline in favor of pure, deterministic physics and multi-agency Model Output Statistics (MOS) consensus. This architectural decision is based on three rigorous engineering imperatives:

- **Absence of Real-Time Ground Truth:** At a 24-hour lookahead horizon (Day T+1), real-time SCADA telemetry actuals are physically unavailable. In the absence of intraday telemetry anchors, generative language models exhibit higher epistemic uncertainty and risk hallucinating sharp hourly ramp gradients that contradict atmospheric thermodynamics.
- **Deterministic Reproducibility Mandate:** Commercial grid dispatch requires 100% mathematical reproducibility. Two independent executions of the Day-Ahead engine on the same weather input must produce identical megawatt commitments down to the nearest kilowatt. Pure numerical ensembling guarantees zero stochastic variance.
- **Ironclad Execution SLA:** The statutory 10:00 AM gate closure is an absolute operational deadline. External API calls to commercial LLM gateways introduce third-party network latency, rate limiting, and potential HTTP timeouts. Eliminating LLM dependencies ensures the Day-Ahead engine completes execution within 3.5 to 5.0 seconds locally on AWS Lambda with an ironclad 99.999% SLA.

1.4 End-to-End Day-Ahead Pipeline Workflow & Orchestration

The Day-Ahead forecasting pipeline is automated via AWS EventBridge cron rules scheduled to execute daily at 09:15 AM IST (45 minutes prior to the statutory 10:00 AM gate closure):

- **Stage 1 — Automated EventBridge Trigger:** AWS EventBridge triggers the containerized AWS Lambda Day-Ahead function via rule cron(45 3 * * ? *) corresponding to 09:15 IST.

- **Stage 2 — Date & Profile Resolution:** The engine resolves execution date D_{today} and target forecast date $D_{\text{target}} = D_{\text{today}} + 1$ day in Asia/Kolkata timezone, loading plant capacity and hardware configuration registries.
- **Stage 3 — Planned Maintenance Ingestion:** The system queries DynamoDB tables (vedanjay-control-windows) to ingest planned maintenance outages, feeder shutdowns, and SLDC export caps.
- **Stage 4 — Deterministic Physics Engines:** High-precision astronomical clear-sky solar ephemeris (solar) or IEC 61400-12 atmospheric air density (wind) is computed for all 96 blocks.
- **Stage 5 — 143-Member NWP Super-Ensembling:** 143 ensemble weather forecast members are fetched across ECMWF IFS, GFS, ICON, and GEM, fitting continuous cubic splines to hourly data.
- **Stage 6 — Power Conversion Physics:** Thermal module derating (solar) or member-by-member non-linear power curve transformation (wind) generates candidate power profiles.
- **Stage 7 — Diurnal Ensembling & Diversity Capping:** Historical 24h-lead-time errors are evaluated; agency diversity capping selects top diverse models; Bayesian inverse-loss weights generate consensus.
- **Stage 8 — Multi-Generator Column Decomposition:** For multi-generator sites (ZTRIC, ENRICH, SHAHA), generation is partitioned across sub-assets and formatted into dedicated columns.
- **Stage 9 — Physical Guardrails & Safety Clamping:** Physical guardrails screen the consensus forecast against night zeroing, clear-sky caps, overcast bounds, and ramp rate limits.
- **Stage 10 — Statutory S3 Data Lake Dispatch:** The finalized 96-block schedule is formatted into statutory CSV files and uploaded to Amazon S3 data lake storage prior to 09:30 AM IST.

2. Part A — Intellis Day-Ahead Solar Power Generation Scheduling System

Part A provides the exhaustive, mathematical, algorithmic, and engineering specification for the Intellis Day-Ahead Solar Power Generation Scheduling System. Operating strictly on pure atmospheric physics and multi-agency Model Output Statistics (MOS) consensus without stochastic LLM dependencies, this engine translates astronomical geometry, satellite-calibrated clear-sky radiance, 143 numerical weather prediction members, dynamic thermal dissipation, planned maintenance schedules, and multi-generator asset topologies into grid-compliant 96-block Day-Ahead schedules.

2.1 Global Variables, Astronomical Constants & System Parameters

The following table defines the global system variables, physical constants, astronomical parameters, and operational coefficients governing the Day-Ahead solar scheduling engine:

Parameter Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
BLOCK_COUNT	N_blocks (int)	Dimensionless	96	Number of standardized 15-minute dispatch intervals per diurnal cycle.
DAY_AHEAD_LEAD_DAYS	Delta_days (int)	Days	1 (Day T+1)	Forecasting lookahead horizon relative to execution date.
STATUTORY_GATE_CLOSURE_IST	T_gate (time)	Time (IST)	10:00:00 IST	Statutory submission deadline under IEGC 2023 grid

				regulations.
SCHEDULE_TRIGGER_IST	T_trigger (time)	Time (IST)	09:15:00 IST	Automated daily AWS EventBridge orchestration trigger time.
SOLAR_CONSTANT	I_0 (float)	W / m ²	1361.2	Extraterrestrial solar irradiance at mean Earth-Sun orbital distance.
STC_REFERENCE_IRRADIANCE	G_STC (float)	W / m ²	1000.0	Standard Test Conditions (STC) reference irradiance for PV module rating.
STC_CELL_TEMPERATURE	T_STC (float)	Celsius	25.0	STC reference photovoltaic cell junction semiconductor temperature.
TEMPERATURE_COEFF_PMP	gamma_pmp (float)	1 / Celsius	-0.0035 to -0.0042	Photovoltaic module maximum power temperature derating coefficient.
FAIMAN_CONVECTION_A	a_faiman (float)	W / (m ² K)	25.0 to 29.0	Faiman thermal convective heat transfer constant for PV module surface.
FAIMAN_CONVECTION_B	b_faiman (float)	W / (m ³ K s)	1.2 to 1.5	Faiman thermal wind convective cooling velocity coefficient.
SANDIA_MODULE_A	a_sandia (float)	Dimensionless	-3.47 to -3.56	Sandia module temperature empirical constant for open-rack ground mount.
SANDIA_MODULE_B	b_sandia (float)	s / m	-0.045 to -0.052	Sandia wind convective cooling empirical exponential rate constant.
ELEVATION_CUTOFF_DEGREES	alpha_min (float)	Degrees	3.0	Solar elevation angle below which solar irradiance is forcibly clamped to 0.0.
SOLAR_WINDOW_START_BLOCK	b_day_start (int)	Index	24 (05:45 IST)	Earliest possible block for non-zero photovoltaic power generation.
SOLAR_WINDOW_END_BLOCK	b_day_end (int)	Index	74 (18:30 IST)	Latest possible block for non-zero photovoltaic power generation.
NWP_ENSEMBLE_MEMBERS	M (int)	Dimensionless	143	Total ensemble weather prediction members ingested across ECMWF, GFS, ICON, GEM.
MAX_MODELS_PER_AGENCY	K_agency (int)	Dimensionless	2	Agency diversity cap: maximum candidate models selected from any single NWP agency.

SLOT_CHAMPION_COUNT	K_slot (int)	Dimensionless	6	Number of diverse champion weather models selected per diurnal operational time slot.
MOS_LOOKBACK_DAYS	N_mos (int)	Days	7	Number of historical days evaluated for 24h-lead-time Model Output Statistics scoring.
MAX_RAMP_FRACTION	R_max (float)	Fraction	0.10 (10% P_ac)	Maximum allowable active power ramp rate change between adjacent 15-minute blocks.

2.2 End-to-End Solar Day-Ahead Execution Flow

At 09:15 AM IST daily, the Intellis Day-Ahead Solar Scheduling System executes the following nine (9) deterministic computational modules in sequential dependency:

- **Module S1: Dynamic Execution Resolution & Plant Configuration:** Resolve execution date D_{today} and target forecast date $D_{target} = D_{today} + 1$ day in Asia/Kolkata timezone; load plant hardware parameters (AC capacity, DC capacity, inverter stations, tilt, azimuth).
- **Module S2: NREL SPA / Ineichen Clear-Sky Ephemeris Engine:** Compute high-precision astronomical ephemeris (solar zenith, azimuth, optical air mass AM); evaluate Ineichen clear-sky radiation triplet (GHI, DNI, DHI); transpose to plane-of-array (POA) via Hay-Davies model; establish deterministic clear-sky ceiling.
- **Module S3: 143-Member NWP Super-Ensemble & Kt Spline Engine:** Ingest 143 ensemble weather members across ECMWF IFS, GFS, ICON, and GEM; compute hourly clearness index $K_t(H)$; fit natural cubic splines; reconstruct continuous 15-minute Global Tilted Irradiance (GTI) matrix.
- **Module S4: Faiman/Sandia Thermal Derating & Base Power Simulation:** Simulate dynamic PV cell temperature $T_{cell}(b)$; compute thermal derate factor $\eta_{thermal}(b)$; apply site empirical transfer multipliers $M[b]$; generate 143 candidate AC generation profiles clamped to AC capacity.
- **Module S5: 24h-Lead-Time MOS Error Accounting & Portfolio Ensembling:** Partition solar day into 3 operational slots (Morning, Peak, Afternoon); score members against ground truth over 7-day historical lookback; enforce agency diversity capping (max 2/agency); compute Bayesian inverse-loss weights; cross-fade slot boundaries.
- **Module S6: DynamoDB Planned Maintenance Outages & Capacity Scaling:** Query DynamoDB table (vedanjay-control-windows) for active maintenance events on target date D_{target} ; parse start/end times into 15-minute blocks; calculate capacity derate ratios for full shutdowns and partial DC derates.
- **Module S7: Multi-Generator Asset Decomposition & Column Allocation:** For multi-generator solar parks (e.g. ZTRIC, ENRICH, SHAHA), allocate total dispatch across sub-assets proportionally; preserve individual asset columns in DataFrames and output CSVs.
- **Module S8: Physical Guardrail Screening & Master Day-Ahead Assembly:** Screen raw predictions against 4 physical guardrails (Night zeroing, Clear-sky ceiling clamp, Overcast optical cloud ceiling, Inverter ramp rate filter); assemble official 96-block schedule vector.
- **Module S9: Statutory 96-Block CSV Formatting & Amazon S3 Dispatch:** Format schedule into statutory CSV files; upload to `s3://{SCHEDULE_BUCKET}/intellis_Dayahead_solar/{SITE}/{TARGET_DATE}/` and raw data lake archives prior to 09:30 AM IST.

2.3 Module S1: Dynamic Execution Resolution, Ingestion & Plant Configuration

Module Context: Day-Ahead scheduling must resolve execution timing and plant topology deterministically. The engine operates strictly in the Indian Standard Time (IST, UTC+05:30) timezone, resolving dates without ambiguity:

$$D_{\text{today}} = \text{date}(\text{now}_{\text{IST}}), D_{\text{target}} = D_{\text{today}} + \Delta_{\text{days}} (\Delta_{\text{days}} = 1)$$

Where: Dynamic calendar date resolution in Asia/Kolkata timezone.

Plant Hardware Profile Ingestion: The registry loads plant-specific metadata, including AC interconnect capacity P_CAP_AC, installed DC nameplate capacity P_CAP_DC, DC-to-AC overloading ratio (ILR), array fixed tilt angle beta, array azimuth angle gamma_m (relative to true south), inverter conversion efficiency eta_inv, and balance-of-system (BOS) loss factor eta_bos.

2.4 Module S2: NREL SPA / Ineichen Clear-Sky Solar Ephemeris & Transposition Engine

Module Context: Theoretical clear-sky radiation represents the physical upper boundary of solar irradiance. The engine evaluates solar ephemeris at each 15-minute block midpoint t_mid(b) for b in {1, 2, ..., 96} using the NREL Solar Position Algorithm (SPA) combined with the Ineichen clear-sky radiation model.

$$\delta = 23.45^\circ \times \sin\left(\frac{360^\circ}{365}(284 + n)\right), \cos(\theta_z) = \sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta)\cos(\omega)$$

Where: Solar declination delta and solar zenith angle theta_z where phi is site latitude and omega is hour angle.

Atmospheric Air Mass & Ineichen Radiation Triplet: Optical air mass AM(b) is computed with barometric pressure altitude correction. The Ineichen model evaluates Direct Normal Irradiance (DNI_cs), Diffuse Horizontal Irradiance (DHI_cs), and Global Horizontal Irradiance (GHI_cs):

$$AM(b) = \frac{1}{\cos(\theta_z) + 0.50572 \cdot (96.07995 - \theta_z)^{-1.6364}} \times \frac{P_{\text{surf}}}{1013.25}, GHI_{\text{cs}}(b) = DNI_{\text{cs}}(b)\cos(\theta_z) + DHI_{\text{cs}}(b)$$

Where: Relative optical air mass and clear-sky horizontal irradiance triplet.

Hay-Davies Plane-of-Array Transposition: The clear-sky irradiance triplet is transposed to the tilted collector plane:

$$POA_{\text{cs}}(b) = DNI_{\text{cs}}\cos(\theta) + DHI_{\text{cs}}\left[A_t \frac{\cos(\theta)}{\cos(\theta_z)} + (1 - A_t)\left(\frac{1 + \cos(\beta)}{2}\right)\right] + GHI_{\text{cs}} \cdot \rho_{\text{albedo}}\left(\frac{1 - \cos(\beta)}{2}\right)$$

Where: Hay-Davies POA transposition where beta is array tilt and theta is angle of incidence.

Horizon Cutoff & Clear-Sky Generation Envelope: If solar elevation $\alpha(b) < 3.0$ degrees (or $\theta_z \geq 87.0$ degrees), $POA_{cs}(b)$ is forcibly set to 0.00 W/m^2 . The theoretical clear-sky electric generation ceiling $P_{clearsky}(b)$ is evaluated as:

$$P_{clearsky}(b) = \min\left(P_{CAP_AC}, P_{CAP_DC} \times \frac{POA_{cs}(b)}{1000.0} \times \eta_{inv} \times \eta_{bos}\right)$$

Where: Clear-sky generation envelope clamped strictly to AC interconnect capacity.

2.5 Module S3: 143-Member NWP Super-Ensemble & Physics-Guided Clearness (Kt) Spline Engine

Module Context: Numerical Weather Prediction (NWP) models provide atmospheric irradiance and temperature forecasts at coarse hourly intervals. Ingesting 143 ensemble members across four major meteorological agencies (ECMWF IFS, GFS, ICON, GEM) provides probabilistic coverage of cloud passage, atmospheric optical thickness, and cloud cover variations.

Mathematical Failure Mode of Direct Linear Interpolation: Directly interpolating hourly irradiance linearly to 15-minute midpoints introduces severe peak-clipping errors of 8% to 15% and distorts morning and afternoon ramp gradients. The Intellis engine solves this through physics-guided Clearness Ratio (Kt) spline reconstruction:

$$K_t(H) = \frac{GHI_{nwp}(H)}{\max(10.0, GHI_{clearsky}(H))}, H \in \{0, 1, \dots, 23\}$$

Where: Atmospheric clearness index K_t computed at hourly synoptic timesteps.

A natural cubic spline is fitted to the 24 hourly clearness ratios $K_t(H)$. High-resolution 15-minute Global Tilted Irradiance for member m is reconstructed by modulating the astronomical clear-sky POA envelope by the continuous spline profile:

$$GTI_m(b) = S_{K_t, m}(t_{mid}(b)) \times POA_{cs}(b), \text{ for all } b \in \{1, \dots, 96\}$$

Where: Physics-guided reconstruction of 15-minute tilted irradiance matrix.

2.6 Module S4: Faiman & Sandia Convective Thermal Model & AC Base Power Generation

Module Context: Photovoltaic module semiconductor conversion efficiency degrades linearly as junction temperature rises. The engine applies the Faiman/Sandia thermal convective model to evaluate dynamic cell temperature for each member m and block b :

$$T_{cell, m}(b) = T_{amb, m}(b) + \frac{GTI_m(b)}{a_{faiman} + b_{faiman} \cdot v_{wind, m}(b)}$$

Where: Dynamic PV cell temperature incorporating ambient temperature and convective wind cooling.

Thermal Power Derating Factor: The temperature derating factor $\eta_{\text{thermal}}(b)$ is computed relative to 25.0 C STC:

$$\eta_{\text{thermal},m}(b) = 1.0 + \gamma_{\text{pmp}}(T_{\text{cell},m}(b) - 25.0), P_{\text{base},m}(b) = \min\left(P_{\text{CAP_AC}}, P_{\text{CAP_DC}} \cdot \frac{\text{GTI}_m(b)}{1000.0} \cdot \eta_{\text{thermal}}\right)$$

Where: Simulated member generation incorporating thermal derating and 96-block transfer multipliers $M[b]$.

2.7 Module S5: 24h-Lead-Time MOS Error Accounting & Portfolio Ensembling

Module Context: Diurnal solar atmospheric dynamics exhibit distinct physical regimes throughout the day. Convective cloud formation, morning haze dissipation, and afternoon monsoonal squalls affect meteorological model accuracy differently. The 96-block solar day is partitioned into three independent operational time slots:

- **Slot 1 — Morning Ramp Window:** Blocks 25 to 40 (06:00 to 10:00 IST): Morning atmospheric boundary layer heating and solar ramp-up.
- **Slot 2 — Peak Solar Window:** Blocks 41 to 56 (10:00 to 14:00 IST): Peak solar irradiation, high cell temperatures, and maximum thermal derating.
- **Slot 3 — Afternoon Decay Window:** Blocks 57 to 74 (14:00 to 18:30 IST): Afternoon solar decay, convective cloud cover, and rapid evening ramp-down.

Candidate Loss Evaluation & Mandatory Agency Diversity Capping: For each candidate member m in $\{1, \dots, 143\}$, the historical Mean Absolute Error (MAE) at a 24-hour lookahead horizon over the past $N_{\text{mos}} = 7$ days is computed for each slot. To eliminate single-agency systematic forecast collapse, an agency diversity cap is strictly enforced: no more than $K_{\text{agency}} = 2$ models may be selected from any single agency (ECMWF, GFS, ICON, GEM). The top $K_{\text{slot}} = 6$ diverse models are selected as slot champions.

$$w_{m,s} = \frac{(\text{MAE}_{m,s})^{-2}}{\sum_{k=1}^{K_{\text{slot}}} (\text{MAE}_{k,s})^{-2}}, P_{\text{slot},s}(b) = \sum_{m=1}^{K_{\text{slot}}} w_{m,s} \cdot P_{\text{base},m}(b)$$

Where: Bayesian inverse-variance loss weighting across slot champion models.

Raised-Cosine Boundary Cross-Fading: To prevent step discontinuities at slot transitions (blocks 40-41 and blocks 56-57), a 4-block raised-cosine blending filter smooths the transition:

$$\alpha(k) = \frac{1}{2} \left[1 - \cos\left(\frac{\pi(k + 0.5)}{L_{\text{blend}}}\right) \right], P_{\text{fused}}(b) = (1 - \alpha)P_{\text{slot},1}(b) + \alpha P_{\text{slot},2}(b)$$

Where: Smooth transition between consecutive diurnal slots across $L_{\text{blend}} = 4$ blocks.

2.8 Module S6: DynamoDB Planned Maintenance Outages & Capacity Scaling

Module Context: Utility solar assets experience planned maintenance shutdowns (inverter servicing, transformer oil filtration, stringCombiner maintenance, or grid curtailments). The engine queries DynamoDB table vedanjay-control-windows for active events scheduled on target date D_{target} .

Capacity Derate Formulation: When an active control window affects block b , the engine derates unconstrained generation:

$$P_{\text{maint}}(b) = P_{\text{fused}}(b) \times \left(1.0 - \frac{\Delta P_{\text{dc_down}}(b)}{P_{\text{CAP_DC}}}\right), P_{\text{curtailed}}(b) = \min(P_{\text{maint}}(b), P_{\text{curtail_ac}}(b))$$

Where: Application of planned DC capacity outages followed by AC export curtailment limits.

2.9 Module S7: Multi-Generator Asset Decomposition & Dedicated Column Allocation

Module Context: Multi-generator complexes (such as ZTRIC, ENRICH, SHAHA) contain distinct sub-generators sharing common grid pooling substations. In Day-Ahead scheduling, statutory SLDC formats require both aggregate plant schedule and individual sub-generator schedule columns in a single official CSV submission.

$$P_{\text{asset},i}(b) = P_{\text{total_plant}}(b) \times \frac{P_{\text{nom_ac},i}}{\sum_{k=1}^{N_{\text{assets}}} P_{\text{nom_ac},k}}$$

Where: Proportional allocation of Day-Ahead generation across individual investor sub-assets.

Multi-Column Output Schema: The resulting DataFrame contains dedicated columns: [Block_No, From, To, TOTAL_MW, ASSET_1_MW, ASSET_2_MW, ..., AvC_MW]. Sub-asset columns reflect individual nominal ratings and asset-specific maintenance status.

2.10 Module S8: Physical Guardrail Screening & Master Day-Ahead Assembly

Module Context: Before official submission, the 96-block consensus schedule is screened against four physical guardrails:

- **GUARDRAIL 1 — Hard Astronomical Night Zeroing:** For all blocks $b < 24$ (before 05:45 IST) and $b > 74$ (after 18:30 IST): $P_{\text{schedule}}(b) = 0.00$ MW strictly. Enforces hard astronomical night zeroing.
- **GUARDRAIL 2 — Clear-Sky Upper Envelope Clamping:** For all daylight blocks b : $P_{\text{schedule}}(b) \leq P_{\text{clearsky}}(b)$. Enforces theoretical clear-sky ceiling.
- **GUARDRAIL 3 — Overcast Optical Cloud Ceiling:** When regional synoptic conditions indicate heavy monsoon overcast ($K_t < 0.35$): $P_{\text{schedule}}(b)$ is clamped at $\max(1.10 * P_{\text{base}}, 0.25 * P_{\text{CAP_AC}})$ to prevent over-optimistic spikes.
- **GUARDRAIL 4 — Ramp-Rate Continuity Filter:** Ramp rate change between adjacent 15-minute blocks is limited to: $\Delta P \leq \max(0.50 \text{ MW}, 0.10 * P_{\text{CAP_AC}})$.

2.11 Module S9: Statutory 96-Block CSV Formatting & Amazon S3 Dispatch

Module Context: Finalized Day-Ahead schedules are formatted into statutory schemas and uploaded via boto3 to Amazon S3 data lake paths prior to 09:30 AM IST (well ahead of the 10:00 AM gate closure):

- **Path 1 — Production Day-Ahead Schedule:** Primary statutory schedule path:
s3://{SCHEDULE_BUCKET}/intellis Dayahead
solar/{SITE}/{TARGET_DATE}/schedule_{SITE}_{TARGET_DATE}_DA.csv
- **Path 2 — Historical Audit Archive:** Archival raw data lake path:
s3://{SCHEDULE_BUCKET}/raw/vedanjay/{SITE}/{TARGET_DATE}/forecasts/strategy2_dayahead_{SITE}_{TARGET_DATE}.csv
- **Path 3 — Governance & Audit Metadata:** Site metadata and model portfolio configuration JSON:
.../dayahead_config_{SITE}_{TARGET_DATE}.json

3. Part B — Intellis Day-Ahead Wind Power Generation Scheduling System

Part B provides the exhaustive, aerodynamic, thermodynamic, and algorithmic implementation specification for the Intellis Day-Ahead Wind Power Generation Scheduling System. Wind power scheduling across a 24-hour horizon (Day T+1) is governed by non-linear cubic kinetic aerodynamics, barometric air density variations, atmospheric boundary layer shear regimes, and wake interference. Unlike solar scheduling, wind generation is continuous across all 96 dispatch blocks (24 hours) and frequently achieves peak generation during nocturnal atmospheric low-level jets.

3.1 Global Variables, Aerodynamic Constants & System Parameters

The following table defines the operational parameters, aerodynamic boundaries, thermodynamic constants, boundary layer shear exponents, and regulatory variables governing the Day-Ahead wind scheduling pipeline:

Parameter Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
GAS_CONSTANT_DRY_AIR	R_spec (float)	J / (kg K)	287.05	Specific gas constant for dry atmospheric air in ideal gas formulation.
STANDARD_AIR_DENSITY	rho_0 (float)	kg / m ³	1.225	Standard sea-level air density specified by IEC 61400-12 power curve normalization.
IEC_DENSITY_EXPONENT	p_density (float)	Dimensionless	1/3 (0.3333)	IEC 61400-12 velocity normalization exponent based on dynamic air density.
CUT_IN_WIND_SPEED	v_in (float)	m / s	3.00	Hub-height wind speed threshold below which aerodynamic rotor torque equals zero.
RATED_WIND_SPEED	vRated (float)	m / s	11.50 to 12.00	Hub-height wind speed at which turbine reaches nameplate

				generator capacity.
CUT_OUT_WIND_SPEED	v_out (float)	m / s	20.00 to 25.00	High-wind emergency aerodynamic pitch-to-feather and mechanical trip threshold.
STORM_RESTART_SPEED	v_restart (float)	m / s	18.00	Hysteresis threshold required for turbine to resume generation following storm cutout.
TOTAL_PARK_DERATE	eta_park (float)	Dimensionless	0.890 (89.0%)	Combined three-factor park derate: wake losses (0.94) x electrical (0.97) x availability (0.98).
SHEAR_EXP_NOCTURNAL	alpha_night (float)	Dimensionless	0.28 to 0.35	Hellmann shear exponent for stable nocturnal boundary layer low-level jet.
SHEAR_EXP_TRANSITION	alpha_trans (float)	Dimensionless	0.18 to 0.22	Hellmann shear exponent during morning boundary layer convective breakdown.
SHEAR_EXP_CONVECTIVE	alpha_day (float)	Dimensionless	0.10 to 0.14	Hellmann shear exponent during strong daytime convective solar heating.
SHEAR_EXP_EVENING	alpha_eve (float)	Dimensionless	0.20 to 0.26	Hellmann shear exponent during evening boundary layer radiative stabilization.
CUBIC_CURVE_EXPONENT	k_power (float)	Dimensionless	2.80	Empirical polynomial exponent for sub-rated wind turbine aerodynamic power curve.
MOS_LOOKBACK_DAYS	N_mos (int)	Days	7	Number of historical days evaluated for 24h-lead-time Model Output Statistics scoring.
DRIVETRAIN_RAMP_MAX	Delta_P_max (float)	MW / block	max(1.0, 0.15*P_rated)	Maximum allowable active power ramp delta to protect gearbox and drivetrain integrity.
CONTINUOUS_WIND_RULE	Rule_24h (bool)	Boolean	True (Enforced)	Strict rule of continuous 24-hour wind operation (strictly NO nocturnal zeroing).
CERC_DSM_TOLERANCE	Tol_cerc (float)	Percentage	+/- 10.0% to 15.0%	Statutory allowable deviation bandwidth under CERC DSM regulations.

3.2 End-to-End Wind Day-Ahead Execution Flow

At 09:15 AM IST daily, the Intellis Day-Ahead Wind Scheduling System executes nine (9) deterministic computational modules in sequential dependency:

- **Module W1: Dynamic Execution Resolution & Hardware Profiling:** Resolve target date $D_{target} = D_{today} + 1 \text{ day}$; ingest fleet hardware profiles (WindTurbineProfile) for Chandawasa (15 MW), JGBPL (30 MW), and Jewli (25.2 MW); establish hub heights, IEC bounds, and capacity limits.
- **Module W2: Dynamic Atmospheric Air Density & Kinetic Flux Engine:** Evaluate barometric surface pressure P_{surf} and temperature T_{2m} for Day $T+1$; compute air density $\rho(b)$; normalize wind speed via IEC 61400-12 density correction.
- **Module W3: 143-Member NWP Super-Ensemble & Boundary Layer Shear:** Ingest 143 ensemble weather members across ECMWF IFS, GFS, ICON, and GEM; extrapolate hub-height wind speeds via Hellmann power-law vertical shear; interpolate via continuous cubic splines.
- **Module W4: Jensen's-Inequality-Safe OEM Aerodynamic Transformation:** Enforce Jensen's inequality proof ($E[v^3] \geq (E[v])^3$); pass EACH candidate wind speed through OEM power curve BEFORE ensembling to prevent 15%-28% under-prediction errors.
- **Module W5: 24h-Lead-Time Wind MOS Evaluation & 4-Regime Ensembling:** Partition 24-hour day into 4 diurnal wind regimes (Nocturnal Jet, Morning Decay, Daytime Lull, Evening Ramp); score 24h-lead-time historical errors; enforce agency diversity capping (max 2/agency); compute Bayesian inverse-loss weights.
- **Module W6: Combined Three-Factor Park Derate & Empirical Multipliers:** Apply combined park efficiency derate $\eta_{park} = 0.890$ (wake, electrical, availability) and 96-block calibrated array transfer multipliers $M[b]$.
- **Module W7: DynamoDB Planned Maintenance & Grid Curtailment Engine:** Query DynamoDB for scheduled turbine maintenance outages; apply turbine availability derate $N_{available} / N_{total}$; cap grid AC export limits.
- **Module W8: Drivetrain Ramp Smoothing & Physical Capacity Clamping:** Verify continuous 24-hour wind operation (NO night zeroing); smooth drivetrain ramp rates ($\Delta_P \leq 15\% \text{ nameplate}$); clamp to interconnect capacity bounds.
- **Module W9: Statutory 96-Block Wind CSV Formatter & Amazon S3 Dispatch:** Format 96-block schedule into canonical 7-column CSV schema; upload to `s3://{SCHEDULE_BUCKET}/intellis Dayhead wind/{SITE}/{TARGET_DATE}/` and raw data lake archives prior to 09:30 AM IST.

3.3 Module W1: Dynamic Execution Resolution & Wind Turbine Hardware Profiling

Module Context: Wind power conversion efficiency is fundamentally governed by wind turbine hardware parameters, aerodynamic operating thresholds, and park-level aerodynamic wake interference. The hardware profile registry manages operational constants across the fleet:

Plant Name	Location / State	Total Nameplate Capacity	Turbine Model / OEM	Hub Height (z_{hub})	Rotor Diameter (D_{rotor})	Turbine Count
CHANDAWASA	Madhya Pradesh	15.0 MW	Suzlon S97 Class IIIA	120.0 m	97.0 m	7 x 2.1 MW

JGBPL	Madhya Pradesh	30.0 MW	Gamesa G97 Class IIIA	104.0 m	97.0 m	15 x 2.0 MW
JEWLI	Maharashtra	25.2 MW	Suzlon S111 Class S	120.0 m	111.0 m	12 x 2.1 MW

Aerodynamic Operating Velocity Limits: Turbines operate within strictly defined wind velocity thresholds: Cut-in velocity $v_{in} = 3.00$ m/s; Rated velocity $v_{rated} = 11.50$ m/s; Cut-out velocity $v_{out} = 20.00$ m/s. During severe atmospheric squalls where v_{hub} exceeds v_{out} , turbines feather blades and trip. Once tripped, generation remains locked at 0.00 MW until wind speeds subside below $v_{restart} = 18.00$ m/s for at least two consecutive blocks (30 minutes).

3.4 Module W2: Dynamic Atmospheric Air Density & Kinetic Wind Flux Engine

Module Context: Aerodynamic rotor lift and mechanical torque are directly proportional to atmospheric air density ρ . Standard OEM turbine power curves assume sea-level air density $\rho_0 = 1.225$ kg/m³. Because inland Indian wind farms (such as Chandawasa and JGBPL in Madhya Pradesh) sit at elevated plateaus (450m to 600m above sea level) with summer ambient temperatures exceeding 42 C, actual air density drops to 1.08 - 1.14 kg/m³. Failing to correct for density causes severe over-prediction errors of 10% to 15%.

$$\rho(b) = \frac{P_{surf}(b) \times 100.0}{R_{spec} \times (T_{2m}(b) + 273.15)}, v_{eff}(b) = v_{hub}(b) \times \left(\frac{\rho(b)}{\rho_0} \right)^{1/3}$$

Where: Ideal gas air density and IEC 61400-12 density-normalized effective wind speed.

A natural cubic spline interpolates hourly air density forecasts to continuous 15-minute midpoint intervals across all 96 blocks.

3.5 Module W3: 143-Member NWP Super-Ensemble & Atmospheric Boundary Layer Shear Engine

Module Context: Standard NWP weather forecasts provide wind speeds at standard 10-meter or 100-meter reference heights. Because modern utility turbines operate at hub heights of 104 meters to 120 meters, wind velocity must be extrapolated vertically through the atmospheric boundary layer using the Hellmann power-law wind shear formulation:

$$v_{hub}(h) = v_{ref}(h) \times \left(\frac{z_{hub}}{z_{ref}} \right)^{\alpha(h)}$$

Where: Hellmann power-law vertical wind shear extrapolation.

Diurnal Atmospheric Boundary Layer Shear Regimes: The shear exponent $\alpha(h)$ fluctuates dynamically across the diurnal heating cycle:

- **Nocturnal Stable Regime ($\alpha = 0.28 - 0.35$):** Regime 1 — Nocturnal Low-Level Jet (21:00 to 06:00 IST): Stable atmospheric stratification decouples vertical mixing, producing strong nocturnal low-level jets with elevated shear: α_{night} in $[0.28, 0.35]$.

- **Morning Transition Regime (alpha = 0.18 - 0.22):** Regime 2 — Morning Boundary Layer Breakdown (06:00 to 10:00 IST): Convective solar heating erodes the nocturnal inversion, triggering downward momentum mixing: alpha_trans in [0.18, 0.22].
- **Daytime Convective Regime (alpha = 0.10 - 0.14):** Regime 3 — Convective Daytime Lull (10:00 to 17:00 IST): Deep vertical atmospheric mixing collapses vertical shear: alpha_day in [0.10, 0.14].
- **Evening Inversion Regime (alpha = 0.20 - 0.26):** Regime 4 — Evening Radiative Stabilization (17:00 to 21:00 IST): Surface radiative cooling terminates convective thermals, accelerating hub-height wind speeds: alpha_eve in [0.20, 0.26].

3.6 Module W4: Jensen's-Inequality-Safe OEM Aerodynamic Transformation Engine

Module Context: Wind power is fundamentally non-linear and cubic in wind speed: P proportional to v^3 . A critical mathematical vulnerability in naive forecasting systems is averaging wind speeds across ensemble members BEFORE passing them through the turbine power curve. By Jensen's Inequality, for any strictly convex function $f(x)$:

$$\mathbb{E}[f(v)] \geq f(\mathbb{E}[v]) \Rightarrow \frac{1}{M} \sum_{m=1}^M v_m^3 \geq \left(\frac{1}{M} \sum_{m=1}^M v_m \right)^3$$

Where: Mathematical proof of Jensen's Inequality for cubic power conversion.

Severe Under-Prediction Penalty: In the steep sub-rated operating zone (5.0 m/s to 9.5 m/s), passing an ensemble-mean wind speed through the power curve underestimates expected electrical power generation by 15% to 28%! The Intellis engine strictly enforces Member-by-Member Non-Linear Power Transformation: EACH candidate member wind velocity $v_{\text{eff},m}(b)$ is individually transformed through the OEM turbine power curve table BEFORE any ensemble averaging or ensembling occurs.

$$P_{\text{gross},m}(b) = f_{\text{OEM}}(v_{\text{eff},m}(b), \rho(b)), P_{\text{net},m}(b) = \min(P_{\text{rated}}, P_{\text{gross},m}(b) \times \eta_{\text{park}} \times M[b])$$

Where: Individual member power curve evaluation followed by park derating and site empirical multipliers.

3.7 Module W5: 24h-Lead-Time Wind MOS Evaluation & 4-Regime Ensembling

Module Context: In Day-Ahead scheduling, wind forecast accuracy varies widely across meteorological agencies depending on the boundary layer stability regime. The 96 blocks are partitioned into four diurnal wind regimes:

- **Slot 1 — Nocturnal Jet Window:** Blocks 1 to 24 (00:00 to 06:00 IST) and Blocks 85 to 96 (21:00 to 24:00 IST): Stable nocturnal low-level jet.
- **Slot 2 — Morning Boundary Decay Window:** Blocks 25 to 40 (06:00 to 10:00 IST): Inversion erosion and downward momentum flux.
- **Slot 3 — Daytime Convective Lull Window:** Blocks 41 to 68 (10:00 to 17:00 IST): Thermal convective mixing and wind speed minimum.

- **Slot 4 — Evening Stabilization Ramp Window:** Blocks 69 to 84 (17:00 to 21:00 IST): Nocturnal stabilization and evening wind ramp.

Champion Selection: For each slot, candidate ensemble members are evaluated against historical 24h-lead-time ground truth MAE. Agency diversity capping restricts selection to a maximum of $K_{\text{agency}} = 2$ models per NWP agency. Bayesian inverse-loss weights are computed, and 4-block raised-cosine blending smooths transitions between regimes.

3.8 Module W6: Combined Three-Factor Park Derate & Empirical Multipliers

Module Context: Combined park efficiency accounts for aerodynamic wake interference, internal medium-voltage collection losses, and balance-of-plant auxiliary consumption:

$$\eta_{\text{park}} = \eta_{\text{wake}} \times \eta_{\text{electrical}} \times \eta_{\text{availability}} = 0.940 \times 0.970 \times 0.980 = 0.890 \text{ (89.0\% Total Park Efficiency)}$$

Where: Three-factor park aerodynamic and electrical efficiency derate.

3.9 Module W7: DynamoDB Planned Maintenance & Grid Curtailment Engine

Module Context: When turbines are scheduled for planned blade inspections, annual servicing, or grid export curtailments, the engine queries DynamoDB table vedanjay-control-windows:

- **Planned Turbine Outage Derate:** When N_{down} turbines are scheduled offline out of N_{total} installed turbines, the available capacity and gross generation are derated by the availability ratio: $P_{\text{derated}}(b) = P_{\text{raw}}(b) * (N_{\text{total}} - N_{\text{down}}) / N_{\text{total}}$.
- **Grid AC Export Curtailment:** When SLDC issues a planned AC export ceiling (e.g. Max 10.0 MW AC), the schedule is capped at 10.0 MW: $P_{\text{schedule}}(b) = \min(P_{\text{derated}}(b), 10.0 \text{ MW})$. The unconstrained generation is preserved in AvC_MW to safeguard commercial deemed generation rights.

3.10 Module W8: Drivetrain Ramp Smoothing, Capacity Clamping & 24h Wind Operation

Module Context: Day-Ahead wind schedules must satisfy strict physical grid code constraints across all 96 blocks:

- **Rule 1 — Continuous 24-Hour Operation:** Strictly NO Solar Night Zeroing: Under no circumstances may nocturnal wind generation be forced to 0.00 MW. Continuous 24-hour generation is preserved across all 96 blocks.
- **Rule 2 — Drivetrain Ramp Smoothing:** Active power transitions between consecutive 15-minute blocks are filtered to $\Delta P \leq \max(1.0 \text{ MW}, 0.15 * P_{\text{rated}})$ to prevent drivetrain stress.
- **Rule 3 — Physical Capacity Clamping:** Schedule values are clamped within physical nameplate bounds: $0.00 \text{ MW} \leq P_{\text{schedule}}(b) \leq P_{\text{rated}}$.

3.11 Module W9: Statutory 96-Block Wind CSV Formatting & Amazon S3 Dispatch

Module Context: Day-Ahead wind schedules are exported to canonical 7-column CSV files: [Block, Time Interval, wind_speed_hub_m_s, air_density_kg_m3, intellis_gti, intellis_mw, schedule_mw] and uploaded via boto3 to:

- **Path 1 — Production Day-Ahead Schedule:** Production schedule path:
s3://{SCHEDULE_BUCKET}/intellis Dayhead
wind/{SITE}/{TARGET_DATE}/schedule_{SITE}_{TARGET_DATE}_DA.csv
- **Path 2 — Historical Audit Archive:** Historical audit path:
s3://{SCHEDULE_BUCKET}/raw/vedanjay/{SITE}/{TARGET_DATE}/forecasts/strategy2_dayahead_{SITE}_{TARGET_DATE}.csv

4. Part C — Integration, Artifact Schema, Edge Cases & Developer Implementation Guide

Part C synthesizes the full operational architecture of the Day-Ahead scheduling system into a concrete engineering reference. It provides the unified data dictionary, AWS Lambda runtime boundaries, a 10-phase developer and intern implementation guide, a matrix of ten (10) critical operational edge cases with automated recovery procedures, and formal architectural sign-off standards.

4.1 Unified Day-Ahead Data Dictionary & Schema Specification

Every file, database record, and data frame produced or consumed by the Day-Ahead system conforms to the following standardized schemas:

Artifact / Object Name	Storage Location / Path	Key Columns / Schema Fields	Update Cadence & Retention
PRODUCTION SOLAR DA CSV	S3: intellis Dayahead solar/{SITE}/{DATE}/schedule_{SITE}_{DATE}_DA.csv	Block_No, From, To, TOTAL_MW, {ASSET_COLS}, AvC_MW	Submitted once daily by 09:30 AM IST. Retained indefinitely.
PRODUCTION WIND DA CSV	S3: intellis Dayhead wind/{SITE}/{DATE}/schedule_{SITE}_{DATE}_DA.csv	Block, Time Interval, wind_speed_hub_m_s, air_density_kg_m3, intellis_gti, intellis_mw, schedule_mw	Submitted once daily by 09:30 AM IST. Retained indefinitely.
AUDIT ARCHIVE CSV	S3: raw/vedanjay/{SITE}/{DATE}/forecasts/strategy2_dayahead_{SITE}_{DATE}.csv	Full 96-block generation profile with internal physics and MOS breakdown	Archived daily for historical accuracy scoring.
DYNAMODB CONTROL WINDOWS	DynamoDB: vedanjay-control-windows	PK: PLANT#EVENT#DATE#TIME, site_id, asset_id, control_type, derate_type, capacity_val, start_time, end_time, is_active	Queried at start of daily Day-Ahead execution.
DAY-AHEAD CONFIG JSON	S3: .../dayahead_config_{SITE}_{DATE}.json	site_id, target_date, P_CAP_AC, P_CAP_DC, selected_models, mos_weights, guardrail_flags	Archived per plant run for compliance auditing.
NWP RAW CACHE CSV	Local / S3: .../weather_nwp_raw_{DATE}.csv	143 ensemble members hourly irradiance, temperature, wind speed, pressure	Cached in /tmp for pipeline execution duration.

4.2 AWS Lambda Execution SLA & Runtime Boundaries

The production Day-Ahead engine runs as an automated containerized AWS Lambda function deployed via Amazon Elastic Container Registry (ECR) on Python 3.11 x86_64 architecture:

- **Memory Allocation:** Memory allocated at 2048 MB to allow parallel vectorized operations via numpy and scipy.
- **Execution SLA & Timeout:** Lambda execution timeout set to 60 seconds. The complete Day-Ahead execution pipeline completes within 3.5 to 5.2 seconds, well below the timeout ceiling.
- **Ephemeral Storage:** Mounted at /tmp with 1024 MB ephemeral disk storage for intermediate CSV caching.
- **Automated Orchestration:** Orchestrated via AWS EventBridge cron rule `cron(45 3 * * ? *)` triggering daily at exactly 09:15 AM IST.

4.3 Step-by-Step Developer & Intern Implementation Guide

To enable an engineering intern or new developer to construct the complete operational Intellis Day-Ahead codebase from scratch, the implementation must follow this precise structural hierarchy and execution sequence:

- **Step 1: Establish Codebase Directory Layout:** Establish modular structure under `schedule/`: `modules/day_ahead_forecast_solar/` (solar ephemeris, Kt splines, Faiman models, MOS selection), `modules/day_ahead_forecast_wind/` (wind profiles, density IEC engine, hub shear, Jensen power curves), and `modules/control_windows/` (DynamoDB maintenance parsers).
- **Step 2: Implement Ephemeris & Transposition Engines:** Implement NREL SPA solar position calculations and Ineichen clear-sky radiation triplet. Implement Hay-Davies POA transposition. Verify that elevation cutoff $\alpha < 3.0$ degrees clamps GTI to 0.0 W/m^2 .
- **Step 3: Implement 143-Member NWP Super-Ensemble Fetcher:** Build Open-Meteo ensemble API connectors fetching 143 members across ECMWF, GFS, ICON, and GEM. Implement hourly clearness ratio $K_t(H)$ and fit natural cubic splines to 15-minute midpoints.
- **Step 4: Implement Thermal & Aerodynamic Conversion Physics:** Implement Faiman/Sandia cell temperature model for solar PV. Implement member-by-member OEM power curve lookup with cubic fallback for wind, strictly preventing Jensen's inequality under-prediction.
- **Step 5: Build 24h-Lead-Time MOS Ensembling Engine:** Implement diurnal slot partitioning. Compute historical 24h-lead-time MAE over past 7 days. Enforce mandatory agency diversity capping (maximum 2 models per NWP agency). Calculate Bayesian inverse-loss weights.
- **Step 6: Implement DynamoDB Planned Maintenance Scaling:** Connect to DynamoDB table `vedanjay-control-windows`. Parse planned outages for target date `D_target`. Implement capacity derates for full shutdowns and partial DC derates.
- **Step 7: Implement Multi-Generator Column Decomposition:** For multi-generator sites (ZTRIC, ENRICH, SHAHA), allocate total generation across sub-assets proportionally. Ensure all sub-generator columns are preserved in output DataFrames.
- **Step 8: Implement Physical Guardrails & Capacity Clamping:** Implement Guardrail 1 (hard night zeroing for solar / continuous 24h operation for wind), Guardrail 2 (clear-sky ceiling clamp), Guardrail 3 (overcast optical ceiling), and Guardrail 4 (ramp rate filter).
- **Step 9: Format Statutory CSV Schemas & S3 Dispatch:** Format schedules into statutory 5-column or 7-column CSVs. Upload to production and raw data lake paths on Amazon S3 via boto3 client.

- **Step 10: Local Unit Testing & End-to-End Replay Verification:** Write unit tests simulating plant runs across clear-sky, overcast, high-wind, and turbine-trip scenarios. Verify execution completes within 5.0 seconds.

4.4 Comprehensive Edge Cases & Automated Failure Mitigation Matrix

The following matrix documents ten (10) critical operational edge cases encountered in Day-Ahead grid scheduling, along with their automated detection triggers, algorithmic responses, and fail-safe recovery procedures:

Edge Case ID	Trigger Condition & Anomaly Description	Immediate Algorithmic Response	Fail-Safe Recovery & Fallback Mechanism
DA-EC-01	Open-Meteo API Network Timeout (No fresh weather feed at 09:15 IST).	Engine detects HTTP connection error or timeout after 10 seconds.	Fallback to most recent 00Z GFS/ECMWF cache stored in S3 raw lake; execute MOS ensembling.
DA-EC-02	DynamoDB Maintenance Overlap (Conflicting shutdown records for D_target).	Module S6/W7 detects overlapping time windows for same asset.	Enforce conservative dominance: Full Shutdown (0 MW) > Partial Derate > AC Export Cap.
DA-EC-03	Extreme Solar Night Generation (Ensemble model predicts non-zero at 02:00 IST).	Guardrail 1 intercepts predictions in blocks $b < 24$ and $b > 74$.	Forcibly clamp $P_{\text{schedule}}(b) = 0.00$ MW strictly. Zero night generation guaranteed.
DA-EC-04	Day-Ahead Severe Storm Cut-Out (Hub wind speed exceeds $v_{\text{out}} = 20.0$ m/s).	Aero engine detects cut-out exceedance; trips turbine power to 0.00 MW.	Maintain 0.00 MW until wind drops below $v_{\text{restart}} = 18.0$ m/s for 2 consecutive blocks.
DA-EC-05	Multi-Generator Sub-Asset Discrepancy (DDB asset_id not in site configuration).	Asset validator catches unrecognized feeder or investor code.	Apply derate at COMBINED plant level proportionally across known assets; alert DevOps.
DA-EC-06	High Ambient Summer Temperature ($T_{\text{amb}} > 45$ C causing extreme thermal derate).	Module S4 evaluates $T_{\text{cell}} > 68$ C; thermal derate factor drops below 0.82.	Verify physical module temperature limits; ensure generation does not fall below physical floor.
DA-EC-07	Dense Monsoonal Overcast False Peak (NWP predicts clear-sky on overcast day).	Guardrail 3 evaluates low $K_t < 0.35$ across majority ensemble members.	Clamp predictions at overcast ceiling: $\max(1.10 * P_{\text{base}}, 0.25 * P_{\text{CAP_AC}})$.
DA-EC-08	Wind Farm Nocturnal Low-Level Jet (Extreme wind shear $\alpha > 0.40$).	Aero engine detects extreme vertical gradient between 10m and 120m.	Cap shear exponent at physical maximum $\alpha_{\text{max}} = 0.35$ to prevent unrealistic generation.
DA-EC-09	Amazon S3 Upload Network Disconnection (S3 endpoint unreachable at 09:25 IST).	boto3 client catches <code>botocore.exceptions.EndpointConnectionError</code> .	Execute 3 exponential retries; cache file to /tmp; trigger secondary SNS alert to ops team.
DA-EC-10	Drivetrain Gearbox Ramp Exceedance ($ \Delta P > 15\%$ rated capacity).	Guardrail 4 identifies sudden multi-megawatt jump between adjacent blocks.	Smooth transition by clamping step change to maximum allowable ramp rate.

5. Architectural Certification and Governance Sign-off

This document constitutes the complete, approved, and authoritative technical specification for the Intellis Day-Ahead Scheduling Architecture (DOC-M04-S01-006, Version v1.0). Any modification to mathematical formulations, physical guardrails, MOS ensembling parameters, or statutory database schemas must be submitted as a formal revision (v1.1+) through the VPPL Engineering Change Control Board (CCB) and recorded in the Master Documentation Registry.